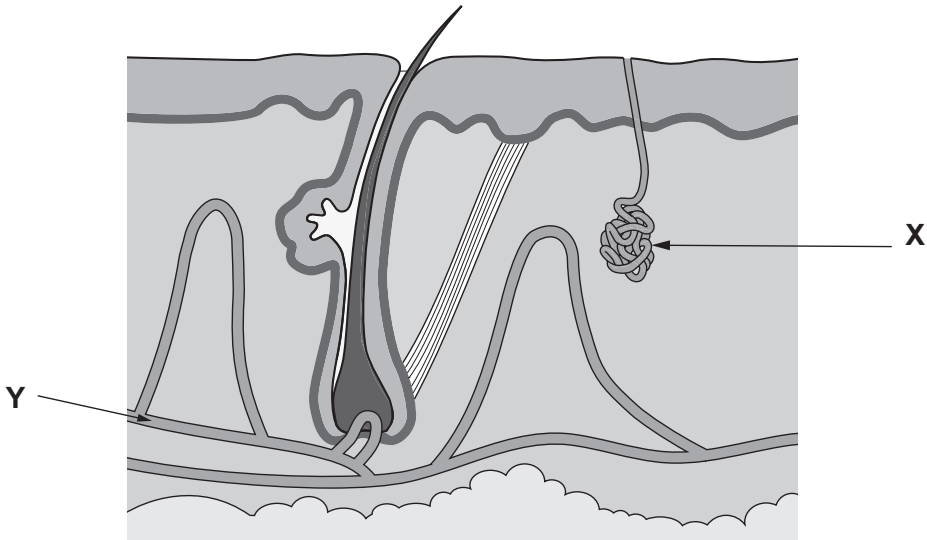


4. (a) The diagram shows a section through the skin.

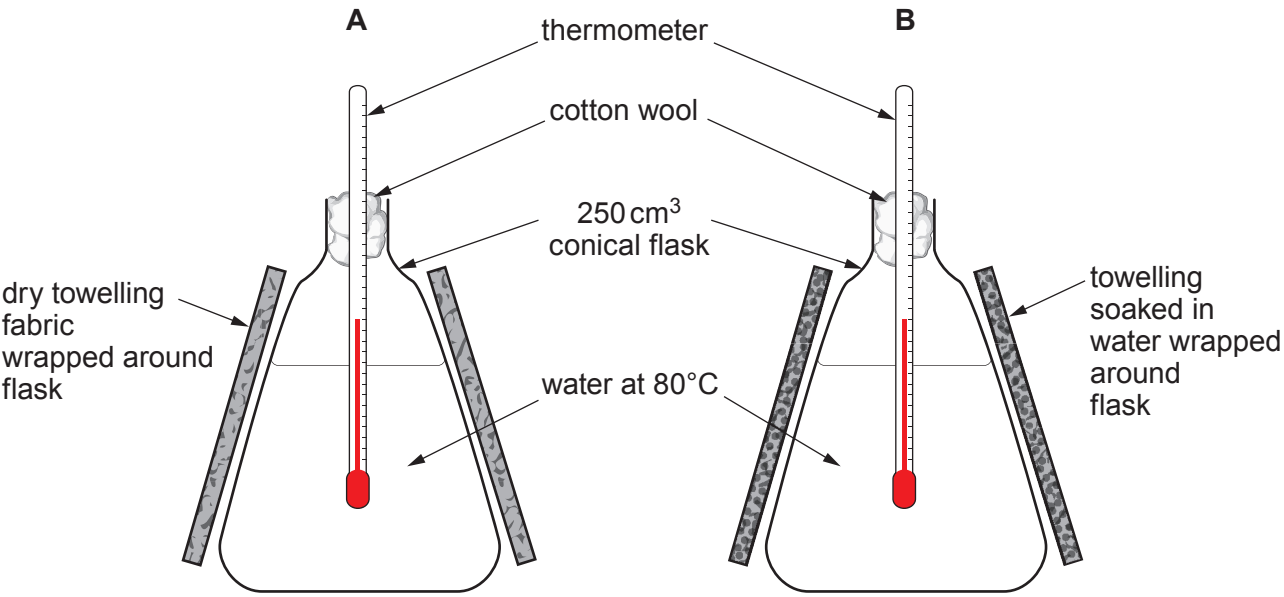


State the name of structures:

(i) X [1]

(ii) Y [1]

(b) Carwyn investigated the control of body temperature.
He set up a model to demonstrate the action of sweat using the apparatus shown below.

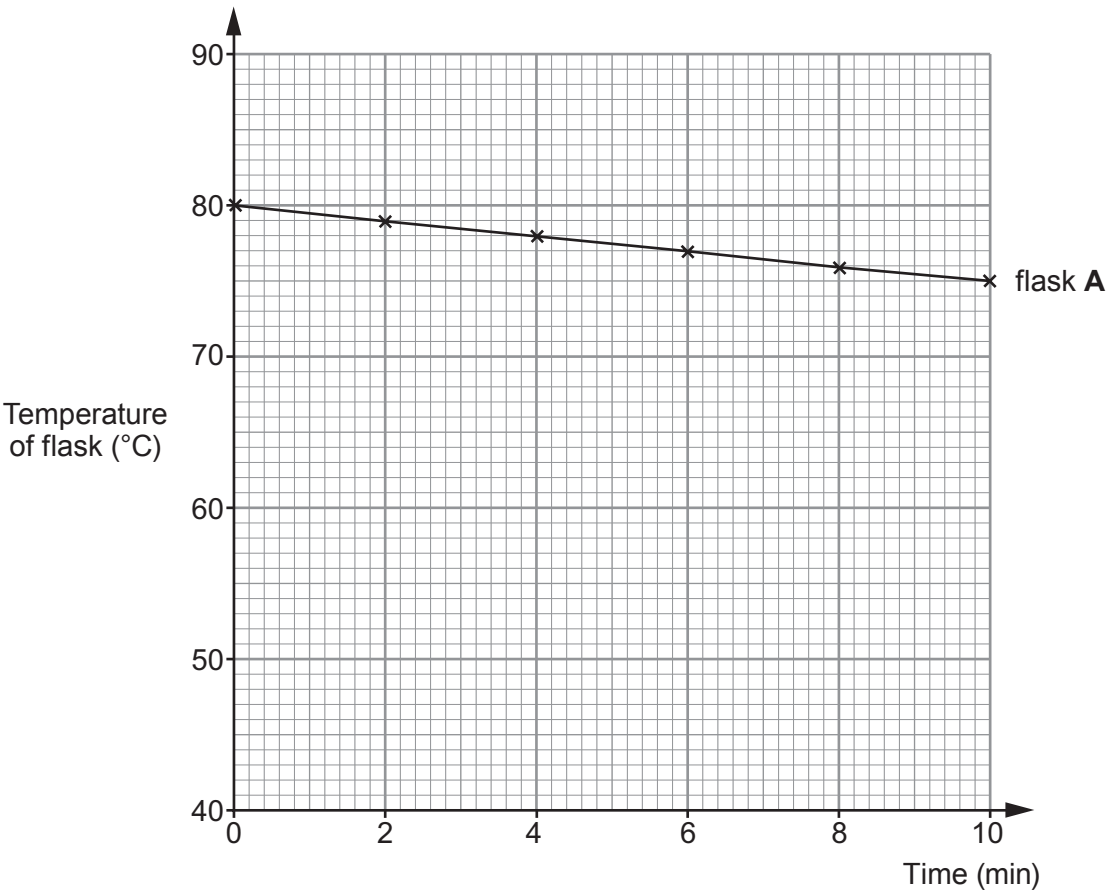


At the start of the investigation, Carwyn added 250 cm³ of water at 80 °C to each flask. He then recorded the temperature of the water in both flasks at two minute intervals for ten minutes, as shown in the results table.

Time (minutes)	Temperature of water in flask (°C)	
	A – dry towelling	B – wet towelling
0	80	80
2	79	73
4	78	65
6	77	58
8	76	54
10	75	51

- (i) **Complete the graph** of the results by:
- I. plotting the points for flask **B**

II. using a ruler to join your plots.
- [3]



Examiner
only

(ii) In Carwyn’s model, the towelling represents the skin surface.
Using the results from flask **B**, explain the effect of sweat on human body temperature. [3]

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(iii) Heat stroke is a condition in which body temperature becomes dangerously high.
Use the results from flask **A** to conclude why people who do not drink enough water during very hot weather may suffer heat stroke. [2]

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Examiner
only

- (i) Use letters **N** and **n** to complete the Punnett square to show how parents **1** and **2** could produce offspring number **4**. [2]

Gametes		

- (ii) State the genotype of individual **9**. Explain your answer. [2]

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- (iii) State the possible genotypes of individual **5**. [1]

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- (c) Our understanding of how characteristics are inherited comes from the principles proposed by Gregor Mendel in the 19th century. In his experiments on pea plants (*Pisum sativum*) Mendel studied characteristics controlled by a single gene. State why this is not representative of most phenotypes. [1]

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